

Tanish Singh Rajpal

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Research focus: reinforcement learning for quantum information systems - routing, control, and algorithm design under noisy, distributed signals

Education

Carnegie Mellon University

M.S. in Information Technology - Information Security

Pittsburgh, PA

Expected May 2027

Mukesh Patel School of Technology Management & Engineering

B.Tech. in Computer Engineering - GPA: 3.83/4.00, Valedictorian

Mumbai, India

Aug 2021 - May 2025

Research Experience

Reinforcement Learning for Heterogeneous Quantum Network Routing

May - Oct 2026

Independent Study, Information Networking Institute, Carnegie Mellon University - Advisor: Prof. Michael Mattarock

- Designed a decentralized PPO routing agent in NetSquid with a fixed, topology-size-invariant observation/action space for reliable decisions from noisy, distributed entanglement-fidelity signals.
- Benchmarked against centralized negative-log-fidelity Dijkstra and hop-count baselines across homogeneous and heterogeneous quantum-hardware noise models.
- Across three training seeds, achieved $\geq 99\%$ mean entanglement fidelity on homogeneous grids from 3×3 to 10×10 and 90-93% under heterogeneous noise; routing success was 100% in five of six configurations and 84.4% $\pm$ 27.0% in the remaining 10×10 homogeneous configuration.
- Validated performance with stochastic evaluation checks to assess robustness beyond deterministic evaluation.

Status: Manuscript in preparation; preprint and code release planned after project completion.

A Hybrid Architecture for Post-Quantum EAP-TLS

Aug 2026 - Apr 2027

Master's Thesis, Information Networking Institute, Carnegie Mellon University - Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock; Reader: Prof. Patrick Tague

- Designing and simulating in ns-3 a hybrid WPA-Enterprise 802.1X architecture combining edge QUIC termination with a QKD-secured backbone to address silent PMTUD blackholes caused by oversized post-quantum ML-DSA EAP-TLS certificates.
- Defined four testable hypotheses spanning handshake latency (targeting $\geq 40\%$ flow-completion-time reduction under elevated wireless error rates), RADIUS compute offload, IPv4/IPv6 baseline isolation, and sustained QKD key-delivery capacity under high-density authentication bursts.

Status: Evaluation in progress; thesis defense and final report expected Spring 2027.

Selected Quantum Computing Project

Quantum Portfolio Optimization Using QAOA - Team Lead

Apr 2026

Yale University YQuantum 2026 Hackathon - 3rd Place | [Repository](#)

- Led a team to a third-place finish by mapping a classical Markowitz portfolio model to a QUBO formulation, converting it to an Ising Hamiltonian, and constructing the corresponding QAOA circuit for insurance portfolio optimization.

Peer-Reviewed Publications

Journal Article

Rajpal, T. S., & Naithani, A. (2026). NeuroCrypt: A Neuro-Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission. *IEEE Transactions on Artificial Intelligence*, 7(1), 512-521. doi:10.1109/TAI.2025.3577605

- Architected a neuro-symbolic framework combining a rule-based cryptographic engine with a deep neural network to dynamically generate symmetric keys; reported a 25% improvement in security resilience and a 15% reduction in key-management overhead against evaluated baselines.

Conference Papers

Naithani, A., & Rajpal, T. S. (2025). Blockchain-Based Digital Health Passport: A Decentralized Framework for Secure and Interoperable Health Data Management. *2025 IEEE Zooming Innovation in Consumer Technologies Conference (ZINC)*, 165-170. doi:10.1109/ZINC65316.2025.11100281

- Co-developed a decentralized health-data architecture using hybrid on/off-chain storage, public-key encryption, and role-based access control.

Naithani, A., Rajpal, T. S., & Mistry, V. (2024). TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout. *2024 International Conference on Advances in Computing, Communication and Materials (ICACCM)*, 1-6. doi:10.1109/ICACCM61117.2024.11059075

- Designed a barcode- and weight-validation smart-cart architecture with UPI payment integration as a lower-cost alternative to RFID-based checkout systems.

Naithani, A., Jain, V., Rajpal, T. S., & Mistry, V. (2024). AI-Based Goalkeeper for Penalty Shot Prediction Using RNN and Pattern Matching Algorithm. *2024 12th International Symposium on Digital Forensics and Security (ISDFS)*, 1-7. doi:10.1109/ISDFS60797.2024.10527307

- Built an RNN and pattern-matching model to predict penalty-kick direction from shooter history; evaluated against recorded 2014-2018 penalty history.

Research Methods & Tools

Methods	PPO, reinforcement learning, network simulation, QAOA, QUBO and Ising formulations
Platforms	NetSquid, ns-3, Qiskit, Git/GitHub
Security & Networking	EAP-TLS, ML-DSA, 802.1X, PMTUD, QKD, Burp Suite, Metasploit

Industry Experience

Newton's Apple - VAPT Intern Oct 2023 - Jan 2024
Pune, India

- Performed penetration testing on a confidential oil-company Android application using Burp Suite and Metasploit, identifying and supporting remediation of three critical vulnerabilities.
- Conducted static Java code analysis and collaborated with developers to remediate recurring security weaknesses in subsequent releases.

Leadership, Awards & Certifications

- **Student Ambassador**, Information Networking Institute, Carnegie Mellon University (Oct 2025 - Present) - primary contact for prospective students and contributor to community-event planning.
- **IBM Qiskit Global Summer School 2026** - completed the full lab series (Labs 0-4), including the Advanced Track; earned the Excellence Badge.
- **1st Place**, Crypto Quest Cybersecurity Competition, Tech-Fest Ambiora (2024).